

FinPilot Scoring and Recommendation Rules

RAG-ready reference document for explaining how FinPilot displays Recommendation, Confidence, and Suggested Allocation.

Document purpose

This document explains the rule-based scoring layer used by FinPilot to convert market research signals into a simple recommendation panel. It is designed to be placed in Amazon S3 and indexed by the MCP-Tools RAG pipeline, so user questions in chat can be answered from this document.

Important: FinPilot is a research aid, not financial advice. The score is a transparent decision-support signal. It does not guarantee returns, does not replace suitability analysis, and does not execute trades without explicit user confirmation.

High-level flow

Step	What Happens	Output Used Later
1	Resolve the user input into a tradable symbol, for example TCS becomes TCS.NS for India or AAPL remains AAPL for the US.	Canonical ticker and market
2	Collect evidence from market data, fundamentals, earnings context, news, and RAG documents.	Normalized evidence bundle
3	Score each signal using deterministic rules. Positive evidence adds points and negative evidence subtracts points.	Component scores
4	Combine the component scores into a total research score from 0 to 100.	Total score
5	Map the total score to Buy, Hold, or Avoid/Sell-style recommendation labels.	Recommendation
6	Estimate confidence based on data coverage, source agreement, and signal strength.	Confidence percentage
7	Convert recommendation and risk controls into a capped suggested allocation.	Suggested allocation percentage

Inputs Used by the Scoring System

Input Group	Examples	Why It Matters
Price performance	3-month trend, recent return, volatility, latest price versus previous close	Captures market momentum and downside risk.
Fundamentals	Market capitalization, P/E, P/B, EPS, ROE, debt to equity, book value, dividend yield	Captures business quality, valuation, and financial strength.
Company profile	Sector, industry, company name, employee count when available	Adds context and helps avoid comparing unlike businesses.
Earnings context	Recent earnings direction, surprises, guidance notes, quarterly commentary	Captures whether fundamentals are improving or weakening.
News context	Recent headlines, business events, regulatory or macro developments	Captures near-term risks and catalysts.

RAG documents	Annual reports, investor presentations, exchange filings, policy documents, internal methodology notes	Provides explainable supporting evidence from PDFs stored in S3.
---------------	--	--

Research Score Formula

The recommendation is based on a weighted total score. Each component is scored on a 0 to 100 scale, then multiplied by its weight. If a component has missing data, FinPilot reduces confidence and uses neutral assumptions instead of pretending the data is known.

```
total_score = (price_score * 0.30) + (fundamental_score * 0.30) + (earnings_score * 0.15) + (news_score * 0.15) + (rag_evidence_score * 0.10)
```

Component	Weight	Positive Signals	Negative Signals
Price score	30%	Uptrend, stable price action, improving relative performance	Sharp drawdown, high volatility, weak recent return
Fundamental score	30%	Reasonable valuation, healthy ROE, positive EPS, manageable leverage	Expensive valuation, weak earnings, high leverage, missing quality metrics
Earnings score	15%	Positive earnings tone, revenue/profit growth, constructive guidance	Earnings misses, weak guidance, falling margins
News score	15%	Positive business updates, product wins, approvals, upgrades	Regulatory risk, litigation, downgrades, adverse macro exposure
RAG evidence score	10%	PDF evidence supports stable business quality or opportunity	PDF evidence highlights material risks or weak outlook

Recommendation Mapping

After calculating the total score, FinPilot maps the score to a recommendation label. The current user-facing recommendation panel is intentionally simple so the investor can scan the result quickly.

Total Score Range	Recommendation	Meaning
75 to 100	Buy	Strong combined evidence. Price, fundamentals, and context are mostly supportive.
55 to 74	Hold	Mixed or moderate evidence. The stock may be reasonable to monitor, but conviction is not high enough for a strong Buy.
40 to 54	Hold / Cautious Hold	Weak or incomplete evidence. Existing holders may review risk, but new allocation should be limited.
0 to 39	Avoid / Sell-style caution	Negative evidence dominates. FinPilot does not present this as personal advice; it flags elevated research risk.

Confidence Calculation

Confidence is not the same as expected return. It describes how reliable the displayed recommendation is, based on evidence completeness and agreement between sources. A Hold with 63% confidence means the system has moderate confidence that Hold is the right research label for the current evidence.

```
confidence = base_confidence + data_coverage_bonus + source_agreement_bonus + signal_strength_bonus - missing_data_penalty - conflict_penalty
```

Factor	Typical Range	How It Affects Confidence
Base confidence	50%	Starting point before evidence quality is considered.
Data coverage bonus	+0% to +15%	Higher when price, fundamentals, news, earnings, and RAG documents are available.
Source agreement bonus	+0% to +10%	Higher when market data, news, and documents point in the same direction.
Signal strength bonus	+0% to +10%	Higher when the total score is clearly far from the Buy/Hold boundary.
Missing data penalty	-0% to -20%	Applied when fundamentals, profile, earnings, or documents are unavailable.
Conflict penalty	-0% to -15%	Applied when price trend is positive but fundamentals/news are negative, or the reverse.

Suggested Allocation Rules

Suggested allocation is a conservative model output for research planning. It is capped by recommendation, confidence, horizon, and risk controls. In the UI example, TCS.NS is rated Hold with 63% confidence and receives a 5.0% suggested allocation for a 3-month horizon and balanced risk profile.

Recommendation	Base Allocation	Adjustment Logic
Buy	8% to 12%	May increase when confidence is high and volatility is controlled.
Hold	3% to 6%	Usually capped near 5% for balanced risk, because evidence is mixed rather than strongly positive.
Cautious Hold	1% to 3%	Used when the system sees weak or incomplete evidence.

Avoid / Sell-style caution	0% to 1%	Used when negative evidence dominates or data quality is too poor.
----------------------------	----------	--

Example: Why TCS.NS Could Display Hold, 63% Confidence, 5.0% Allocation

Displayed Field	Example Value	Explanation
Recommendation	Hold	The total score sits in the Hold range because the evidence is useful but not strong enough for Buy.
Confidence	63%	The system has moderate confidence. Some evidence is available, but confidence is reduced if detailed fundamentals, earnings, or document context are incomplete.
Suggested Allocation	5.0%	A Hold label with balanced risk and a 3-month horizon is capped near 5%. This keeps exposure moderate.
Explanation text	For a 3 months horizon...	The text summarizes the recommendation, horizon, risk profile, and evidence categories used.

How Chat Should Use This Document Through RAG

When a user asks questions such as "Why did FinPilot recommend Hold?", "How is confidence calculated?", or "Why is suggested allocation 5%?", the chat backend should call the MCP RAG search tool. The RAG tool searches PDFs stored in S3, retrieves relevant chunks from OpenSearch Serverless, and returns evidence to the FastAPI agent. Streamlit should only display the final answer.

Recommended S3 location

Upload this file to:
s3://dstrmaysam-finpilot/documents/methodology/finpilot_scoring_and_recommendation_rules.pdf

Suggested retrieval queries

- How does FinPilot calculate recommendation?
- Why is the recommendation Hold?
- How is confidence percentage calculated?
- How does FinPilot calculate suggested allocation?
- What inputs are used in the FinPilot rule-based scoring system?
- Why does missing fundamentals data reduce confidence?

Required behavior for the backend agent

Requirement	Expected Behavior
Use MCP RAG tool	The FastAPI backend calls finpilot_rag_search_documents or finpilot_search_documents before answering methodology questions.
Cite retrieved evidence	The answer should mention that it is based on the FinPilot scoring methodology document when that document is retrieved.
No frontend processing	Streamlit sends the user question to FastAPI and renders the response. It should not perform retrieval, scoring, or LLM reasoning.
Graceful fallback	If RAG returns no evidence, the backend should say the methodology document was not found in the indexed knowledge base.

Limitations and Guardrails

- The scoring system is deterministic and explainable, but it is only as good as the available data.
- Missing Indian stock fundamentals, incomplete earnings data, or unavailable RAG documents lower confidence.
- Recommendation labels are research labels, not personal financial advice.
- Suggested allocation is capped and conservative; it does not know the user's full financial situation.
- The system must not promise guaranteed profit, must not encourage over-concentration, and must require user confirmation before any trade execution.

Short Answer Template for Chat

If a user asks why FinPilot displayed Hold, 63% confidence, and 5.0% suggested allocation, the assistant can answer:

FinPilot maps evidence into a weighted research score. Price performance and fundamentals each carry 30%, earnings and news each carry 15%, and RAG document evidence carries 10%. A Hold appears when the total score is moderate or mixed rather than strongly

positive. Confidence reflects data coverage, source agreement, and signal strength, with penalties for missing or conflicting evidence. Suggested allocation is then capped by recommendation, confidence, investment horizon, and risk profile. For a Hold with balanced risk and a 3-month horizon, the allocation is usually kept around 3% to 6%, so 5.0% is a conservative midpoint.